

Kinetics of Substrate-Limited Growth in a Stirred Batch Reactor

A. Hassan, M. Saleh and R. Farouk

Department of Biotechnology, Faculty of Science

Abstract. We measure the specific growth rate of a yeast culture on glucose and fit the Monod model to twelve batch runs. The maximum rate is 0.42 per hour and the half-saturation constant is 0.18 grams per litre. The yield of biomass on substrate stays close to 0.5 over the whole range of initial concentrations, which supports a constant-yield model for design.

$$Y_{X/S} = fX / fS$$

1. Introduction

Batch cultures are the simplest way to study how cells grow on a single limiting substrate. During exponential growth the biomass concentration rises at a rate proportional to itself, and the constant of proportionality is the specific growth rate. When the substrate becomes scarce, the rate falls and the culture enters the stationary phase.

Monod proposed that the specific growth rate depends on the substrate concentration in the same way that an enzyme rate depends on its substrate. This simple model still describes many cultures well and is the starting point of most reactor designs.

2. Model

The specific growth rate follows the Monod equation:

$$\mu = \mu_{max} S / (K_S + S)$$

and the biomass grows exponentially while the substrate is in excess:

$$X(t) = X_0 e^{\mu t}$$

The yield coefficient links the two balances, f marking the observed value:

3. Methods

Twelve runs were made in a 2 L stirred vessel at 30 °C and pH 5.0, with initial glucose from 1 to 20 g/L. Samples were taken every hour; biomass was measured as dry weight and glucose by an enzymatic assay. Rates were found from the slope of the logarithm of biomass against time.

4. Results

Table 1 lists the fitted parameters. The fit is good over the whole range, and the residuals show no trend with the initial concentration.

Parameter	Value	Unit
μ_{max}	0.42	h^{-1}
K_S	0.18	g/L
$Y_{X/S}$	0.51	g/g

The doubling time at the maximum rate is $\ln 2 / \mu_{max} = 1.65$ h, in line with earlier reports for the same strain.

5. Conclusion

A constant yield and Monod kinetics describe these cultures well enough for the design of a batch process.